Transverse & Longitudinal Waves

Exam code: 0625 & 0972

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Transverse waves

- Waves can exist as one of two types:
  - **Transverse**
  - **Longitudinal**
- Transverse waves are defined as:

Waves where the direction of vibration is at right angles to the direction of energy transfer

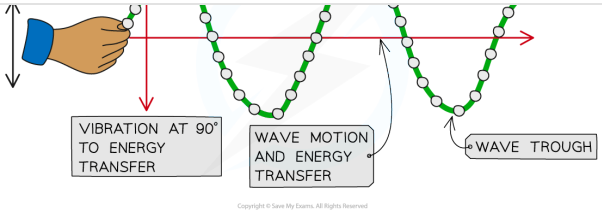
- For a transverse wave, the oscillation is **perpendicular to** the direction the wave is travelling in
- Mechanical transverse waves can move in **solids**, and on the surface of liquids, but not in liquids or gases
- Non-mechanical transverse waves can move in a **vacuum**

Transverse wave motion

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



Transverse waves can be seen in a rope when it is moved quickly up and down

- Examples of waves that can be modelled as transverse are:
  - Electromagnetic waves (such as radiowaves, visible light, X-rays etc)
  - Ripples on the surface of water
  - Seismic S-waves (secondary earthquake waves)

Longitudinal waves

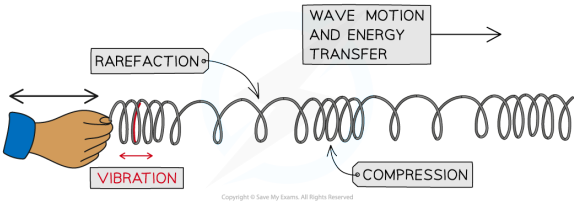
- Longitudinal waves are defined as:

Waves where the direction of vibration is parallel to the direction of propagation

- For a longitudinal wave:
  - The energy transfer is in the **same direction** as the wave motion
  - They can move in solids, liquids and gases
  - They can **not** move in a vacuum (since there are no particles)

- The key features of a longitudinal wave are where the points are:
  - Close together, called **compressions**
  - Spaced apart, called **rarefactions**

Longitudinal wave motion



Longitudinal waves can be seen in a slinky spring when it is moved quickly backwards and forwards

- Examples of waves that can be modelled as longitudinal waves are:
  - Sound waves
  - Seismic P-waves (primary earthquake waves)

Comparing transverse and longitudinal waves

Property	Transverse waves	Longitudinal waves
Structure	Peaks and troughs	Compressions and rarefactions
Vibration	Right angles to the direction of energy transfer	Parallel to the direction of energy transfer
Vacuum	Only electromagnetic waves can travel in a vacuum	Cannot travel in a vacuum
Material	Can move in solids and the surfaces of liquids	Can move in solids, liquids and gases
Density	A constant density	The density of the wave changes
Pressure	Has a constant pressure	Pressure in the wave changes
Speed of wave	Depends on the material the wave is travelling in	Depends on the material the wave is travelling in

Examiner Tips and Tricks

The key difference between transverse and longitudinal waves is the direction of the vibrations with respect to the direction of the wave itself. For transverse waves, these are perpendicular to each other, whilst for longitudinal waves, these are parallel.

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